



Python Conditional Logic

Data with Dylan | Guided Notes | Python Fundamentals #2

Name: _____

Date: _____

- ☒ **HOW TO USE THESE NOTES:** Watch the video and fill in the blanks as Dylan explains each concept. Use the writing boxes to capture your own notes and examples. Complete the practice problem at the end!

Section 1: User Input

Topic: Letting a user interact with a Python program

What is user input?

Complete the definition:

The `input()` function lets a user type _____ after seeing a _____.

`input()` is a built-in Python _____.

The value returned by `input()` is always a _____.

Complete the code below:

```
name = _____("Enter in your name: ")
print("My name is", _____)
```

What will the program display after the user enters their name?

Getting number input

When you want the user to enter a whole number, wrap `input()` inside the _____ function.

```
age = _____(input("Enter in your age: "))
```

Why is this conversion needed?

If a user types "Dylan" when the program expects an integer, Python raises an error because the text cannot be converted into an integer.

In your own words - how does user input make a Python program interactive?

Section 2: Conditions & Comparison Operators

Topic: Checking whether an expression is True or False

What is a condition?

Complete the definition:

A condition is an expression that evaluates to a _____.

A boolean can only be one of two values: _____ or _____.

Conditional logic helps a program decide which _____ to run based on a condition.

Comparison Operators

CODE	OPERATOR NAME	RESULT / NOTES
>	_____	True when the left value is _____ the right value.
<	_____	True when the left value is _____ the right value.
>=	_____	True when the left value is greater than or _____ the right value.
<=	_____	True when the left value is less than or _____ the right value.
==	_____	Checks whether two values are _____.
!=	_____	Checks whether two values are _____.

Predict the results

EXPRESSION	RESULT
5 == 5	_____
5.0 == 5	_____
"5" == 5	_____
"5" != 5	_____
10 + 5 > 20	_____
10 + 5 <= 15	_____

Circle the VALID condition

Which line correctly checks whether num % 2 is equal to 0?

Choice A

```
if num % 2 = 0:
```

Choice B

```
if num % 2 == 0:
```

Why is the other choice incorrect?

Section 3: if Statements

Topic: Running code only when a condition is True

How does an `if` statement work?

Complete the definition:

An `if` statement checks a _____.

The indented block beneath an `if` statement runs only when the condition is _____.

```
if _____:  
    _____
```

Important syntax rules

Every `if` condition ends with a trailing _____.

The code inside the conditional block must be _____.

In your own words - what does indentation tell Python?

What does the trailing colon `:` tell Python?

Trace the path

CONDITION RESULT	WHAT HAPPENS TO THE INDENTED CODE?
True	_____
False	_____

Section 4: `elif` & `else` Paths

Topic: Handling more than one possible outcome

Complete the definitions

elif stands for _____.

An elif condition is checked only after the previous condition is _____.

An else block runs when all earlier conditions are _____.

Conditional path structure

```
if condition_1:
    # runs when condition_1 is True

elif condition_2:
    # runs when condition_1 is False and condition_2 is True

else:
    # runs when all previous conditions are False
```

Python checks conditional paths from _____ to _____.

In your own words - why can an else block make code cleaner than writing one more elif condition?

Logical Operator Reference

Reference note: Track these operators as a future-study extension.

CODE	OPERATOR NAME	RESULT / NOTES
and	_____	_____
or	_____	_____
not	_____	_____

Section 5: Example Program - Even or Odd

Topic: Combining user input, comparison operators, and conditional paths

Program goal

Ask the user for a number, reject negative input, and determine whether the number is even or odd.

```
num = int(input("Please enter in a positive number: "))
```

Complete the conditional logic

```

if num < ____:
    print("You entered an invalid number")

elif num % 2 == ____:
    print("Your number was even")

else:
    print("Your number was odd")

```

Why does this work?

The modulo operator % returns the _____ after division.

A number is even when dividing it by 2 leaves a remainder of _____.

Why can the final odd-number path use else?

Trace the program

USER INPUT	FIRST CONDITION: num < 0	NEXT PATH / OUTPUT
15	_____	_____
-10	_____	_____
8	_____	_____

Section 6: Number Guessing Game & Nested if Statements

Topic: Using conditions inside other conditions

Setting up the game

```

import random

secret = random.randint(1, 10)
guess = int(input("Please enter a number between 1 and 10: "))

```

random is a Python _____.

random.randint(1, 10) creates a random _____ between
_____ and _____.

First-guess logic

CONDITION	PROGRAM RESPONSE
guess == secret	_____
guess < secret	_____

CONDITION	PROGRAM RESPONSE
Otherwise	

Nested conditions

Complete the definition:

A nested if statement is an if statement placed _____ another conditional block.

Why must the nested code be indented farther than the outer conditional?

Complete the nested structure

```
if guess == secret:
    print("You got it right")

elif guess < secret:
    print("Your number was too low")
    guess_again = int(input("Guess again: "))

    if guess_again == secret:
        print("You got it right")
    else:
        print("You got it wrong. The number was", secret)

else:
    print("Your number was too high")
    _____
```

Section 7: Putting It All Together

Topic: Conditional logic reference table

Using what you have learned, complete the full conditional logic reference table:

TYPE / CONCEPT	WHAT IT STORES	EXAMPLE	YOUR OWN EXAMPLE
input()	A user's typed response as a string	name = input("Enter your name: ")	
int(input())	A typed response converted to an integer	age = int(input("Enter your age: "))	
Condition	An expression that evaluates to True or False	score >= 70	
if	The first conditional path to check	if score >= 70:	
elif	An additional condition checked after earlier conditions are False	elif score >= 60:	
else	The remaining path when earlier conditions are False	else:	

TYPE / CONCEPT	WHAT IT STORES	EXAMPLE	YOUR OWN EXAMPLE
Nested if	A conditional inside another conditional block	<code>if guess < secret:</code>	
<code>random.randint()</code>	A random integer within a chosen range	<code>random.randint(1, 10)</code>	

Section 8: Quick Recap

Test yourself - try to complete these without looking back!

1. The `input()` function always returns a _____.
2. To convert a user's typed response into an integer, use _____.
3. A condition evaluates to either _____ or _____.
4. Use _____ to check whether two values are equal, but use _____ to assign a value to a variable.
5. The indented code beneath an `if` statement runs only when the condition is _____.

Practice Problem

 **Challenge - Pause the video and try this yourself before Dylan reveals the answer!**

```
import random
secret = random.randint(1, 10)    guess = int(input("Please enter a number between 1 and 10: "))
```

Goal: Build the conditional logic for a number guessing game with only two tries.

- If the first guess is correct, print "You got it right".
- If the first guess is too low or too high, print a hint and ask for one more guess.
- After the second guess, print whether the user got it right or wrong. Reveal the secret number when they are wrong.

Why must both guesses use `int(input(...))` instead of only `input(...)`?

Why does the second-guess code need to be nested inside the "too low" and "too high" paths?

Write your full statement below:

How did it go? What part tripped you up?

Personal Notes

Use this space to capture any additional observations, questions, or ideas from the video.

Key Terms Reference

Use this table to build your Python vocabulary. Write definitions in your own words.

TERM	DEFINITION IN YOUR OWN WORDS
Conditional Logic	
Condition	
Boolean	
True	
False	
User Input	
Prompt	
Function	
Built-in Function	
input()	
int()	
Integer (int)	
String (str)	
Comparison Operator	
>	
<	
>=	
<=	
==	
!=	

Key Terms Reference (continued)

Continue building your Python vocabulary. Write definitions in your own words.

TERM	DEFINITION IN YOUR OWN WORDS
Assignment Operator (=)	
Indentation	
Trailing Colon (:)	
if	
elif	
else	
Nested if Statement	
Modulo (%)	
Module	
import	
random	
random.randint()	
and	
or	
not	
Invalid literal error	